Q-1 Definition and concept of Disasters and Disaster Management Cycle(phases)

- Concept

- Disaster Management is the planned steps taken to minimize the effects of a disaster, and to be able to proceed to business continuity stage.

– Disaster Management includes sum total of all activities, programme and measures which can be taken up before, during and after a disaster with the purpose of avoiding, reducing the impact or recovering from its losses.

– Disaster Management means managing resources and various responsibilities to deal with all humanitarian aspects or emergencies.

-- This may include preparedness before disaster, response and recovery i.e. rebuilding and supporting society.

-- The purpose of this is to lessen the impact of disasters.

*** Aims of Disaster Management**

- The aims of Disaster Management are:

- To reduce the potential losses from hazards (avoid, if possible).
- To assure prompt and appropriate assistance to victims when necessary.
- To achieve rapid and durable recovery.

*** Importance of Disaster Management**

- Over past 20 years disasters have affected 4.4 billion people, caused $2 trillion of damage and killed 1.3 million people.

- These losses have outstripped the total value of official development assistance in the same period.

- Natural disasters disproportionately affect people living in developing countries and the most vulnerable communities within those countries.

- At the global level, there has been considerable concern over natural disasters.

- Even as scientific and material progress is made, the loss of lives and property due to disasters has not decision.

- India has been traditionally vulnerable to natural disasters on account of its unique geo-climate conditions.

- Floods, droughts, cyclones, earthquakes and landslides have been recurrent phenomena.

*** Phases of Disaster Management (Disaster Management Cycle)**

- Disaster Management activities can be grouped into five phases that are related by time and function to all types of emergencies and disasters.

- These phases are also related to each other, and each involves different types of skills.

- These phases are:

- Planning and Mitigation
- Preparedness
- Response
- Recovery

*** Planning Phase**

- Activities necessary to analyse and document, the possibility of an emergency or disaster and the potential consequences or impacts on life, property, and the environment.

- This includes assessing the hazards, risks, mitigation, preparedness, response, and recovery needs.

*** Mitigation Phase**

- Activities that actually eliminate or reduce the probability of a disaster (e.g. arms build-up to deter enemy attack).

- It also includes long-term activities designed to reduce the effects of unavoidable disaster (e.g. land use management, establishing comprehensive emergency management programs such as vegetation, building restrictions in potential flood zones).

*** Preparedness Phase**

- Activities necessary to the extent that mitigation measures have not, or cannot, prevent disasters.

- In the preparedness phase, governments, organizations, and individuals develop plans to save lives and minimize disaster damage (e.g. compiling state resource inventories, installing early warning systems, and preparing predetermined emergency response forces).

*** Response Phase**

- Activities following an emergency or disaster.

- These activities are designed to provide emergency assistance for victims (e.g. search and rescue, emergency shelter, medical care, and mass feeding).

- They also seek to stabilize the situation and reduce the probability of secondary damage (e.g. shutting off contaminated water supply sources, and securing and patrolling areas prone to looting) and to speed recovery operations (e.g. damage assessment).

*** Recovery Phase**

- Activities necessary to return all systems to normal or better.

- They include two sets of activities:

➢ Short-term recovery activities return vital life-support systems to minimum operating standards (e.g. clean-up, temporary housing, and access to food and water).

➢ Long-term recovery activities may continue for a number of years after a disaster. Their purpose is to return life to normal or improved levels (e.g. redevelopment loans, legal assistance, and community planning).

• Classification of Disasters

- A common way to classify hazards is by category.

1) Geological Disasters
➢ Earthquakes, Landslides, Tsunami and mining, etc.
2) Hydro-Meteorological Disasters
➢ Floods, Cyclones, Lightening, Thunder-storms, Hail storms, Avalanches, Droughts, Cold and heat waves.
3) Biological Disasters
➢ Epidemics, Pest attacks, Forest fire etc.
4) Technological Disasters
➢ Chemical, Industrial, Radiological, Nuclear, etc.
5) Manmade Disasters
➢ Building collapse, Rural and urban fire, road and rail accidents, etc.

Q-2. Hazard

- Concept

- A **hazard** is a situation that poses a level of threat to life, health, property, or environment.

- A dangerous phenomenon, substance, human activity or condition that may cause loss of life, injury or other health impacts, property damage, loss of livelihoods and services, social and economic disruption, or environmental damage hazards are conditions that have the potential to harm to a community or environment.

*** Types of Hazard**

- A common way to classify hazards is by category.

1) Biological Hazard
➢ Bacteria, viruses, insects, plants, birds, animals, and humans, etc.
2) Chemical Hazard
➢ Depends on the physical, chemical and toxic properties of the chemical.
3) Ergonomic Hazard
➢ Repetitive movements, improper set up of workstation, etc.
4) Physical Hazard

- Radiation, magnetic fields, high pressures, noise, etc.

5) Psychosocial Hazard

- Stress, violence, etc.

6) Biological Hazard

- Slipping/tripping, equipment malfunctions or breakdowns, etc

Q-3 Vulnerability

- Concept

- In relation to hazards and disasters, vulnerability is a concept that links the relationship that people have with their environment to social forces and institutions and the cultural values that sustain and contest them.

- Exposed to the possibility of being attacked or harmed, either physically or emotionally.

- Vulnerability refers to the inability to withstand the effects of a hostile environment

Q-4. Risk

- Concept

- Risk is the potential of losing something of value.

- The values such as physical health, social status, emotional well-being or financial wealth.

- A probability or threat of damage, injury, liability, loss, or any other negative occurrence that is caused by external or internal vulnerabilities.

– Risk is an uncertain event or condition that, if it occurs, has an effect on at least one objective.

- Risk means 'effect of uncertainty on objectives'.

- What is Risk Assessment?

- Risk Assessment is the process where you:

- Identify hazards,
- Analyse or evaluate the risk associated with that hazard,
- Determine appropriate ways to eliminate or control the hazard, and
- Review the ways of elimination or control.

Q-5 Foundation of Emergency Management

- All phases of emergency management depend on data from a variety of sources.

- The appropriate data has to be gathered, organized, and displayed logically to determine the size and scope of emergency management programs.

- During an actual emergency it is critical to have the right data, at the right time, displayed logically, to respond and take appropriate action. Emergencies can impact all or a number of government departments.

- Emergency personnel often need detailed information concerning pipelines, building layout, electrical distribution, sewer systems and so forth.

- By utilizing a GIS, all departments can share information through database on computer-generated maps in one location.

- Most emergencies do not allow time to gather these resources. This results in emergency responders having to guess, estimate, or make decisions without adequate information.

- Most of the data requirements for emergency management are of a spatial nature and can be located on a map.

Q-6 Earthquake

- Earthquakes are the deadliest of all natural disasters.

- Most deaths are caused by collapsing buildings or fires.

* Causes

- Earthquakes occur within the Earth's crust along faults that suddenly release large amounts of energy that have built up over long periods of time.

- The shaking during an earthquake is caused by seismic waves.

- The earth's crust near tectonic plate edges are forced to bend, compress, and stretch due to the internal forces within the earth, causing earthquakes.

- The continents and oceans rest on top of giant slabs called TECTONIC PLATES which make up earth's outer crust. These plates float like rafts on the hot, semi-liquid mantle below the crust.

- Most earthquakes begin deep underground at a point called the focus. As the rocks shatter at the focus, shake waves called seismic waves radiate outward in all directions.

* Causes of Earthquake

- Economic activities like agriculture, industry, trade and transport are severely affected.
- Water pipes, sewers are disrupted
– Communications such as telephone wires are damaged.
- Chances of fire short-circuit.
- Chances of floods.
- Develop cracks in dams.
- Damage to infrastructure like roads, electric poles, telephone lines etc.
- Landslides shaking and ground rapture.
- Fires soil liquefaction.
- Tsunami floods in ocean.

* Prevention and Protection from Earthquake

- If you are in house
  - Don't use lift.
  - Be prepared to move with your family.
- If you are in shop, school or office
  - Don't run for an exit.
  - Take cover under a disk/table.
  - Move away from window glass.
  - Do not go near electric point and cable.
- If you are outside
  - Avoid high buildings, walls, power lines and other objects that could fall and create block.
  - Don't run through streets.
  - If possible, move on to an open area away from hazard including trees.
- If you are in vehicle
  - Stop in a safe open place.
  - Remain inside vehicle.
  - Close window, doors and vents.
- Keep calm, switch on the transistor radio and obey instructions.
- Keep away from beaches and low bank of river.
- Turn-off the water, gas and electricity.
- Do not smoke, light match or use a cigarette lighter.

Q-7 Landslide

- A landslide is an catastrophic event, causing significant loss of life and money.
- It is a gravity slope failures which are triggered by earthquake, ground shaking or excess precipitation.
- It is a downward and outward movement of slope forming materials composed of rocks, soils, artificial fills or combination of all these materials along surfaces of separation by falling, sliding and flowing, either slowly or quickly from one place to another.

*** Factors causing landslides:**

- Erosion
– Precipitation
- Vegetation
- Soil
- Earthquakes
- Volcanic eruptions
- Human activities such as deforestation, vibrations from machinery or blasting

*** Types of Landslides**

@ Falls
- Abrupt movements of materials that become detached from steep slopes or cliffs, moving by free-fall, bouncing, and rolling.

@ Flows
- Many types of mass movement such as creep, debris flow, debris avalanche, and mudflow.

@ Creep
- Slow, steady downslope movement of soil or rock, tilted poles or fences.

@ Debris flow
- Rapid mass movement in which loose soils, rocks and organic matter combine with entrained air and water.

@ Debris avalanche
- A variety of very rapid to extremely rapid debris flow.

@ Mudflow
- Rapidly flowing mass of wet material that contains at least 50% sand, silt and clay sized particles.

@ Lateral spreads
- Often occur on very gentle slopes and result in nearly horizontal movement of earth materials.

Q-8 Tsunami

- Tsunami is Japanese word and means “Harbour Wave”.
- Tsunami are water waves or seismic sea waves caused by large-scale sudden movement of the sea floor.
- Tsunami can cause serious damage thousands of kilometres from the causative faults.
- Tsunami travel at incredible speeds, as much as 900 km/hr, and the distance between wave crests can be as much as 500 km.
- As the waves approach shallow water, a tsunami’s speed decreases and the energy is transformed into wave height, sometimes reaching as high as 25 m.

*** Plans for a Tsunami**

- Develop a family disaster plan.
- Learn about tsunami risk in your community.
- If you are visiting an area at risk from tsunamis, check with the hotel, motel, or campground operators for tsunami evacuation information.
- Plan an evacuation route from your home, school, workplace, or any other place you will be where tsunamis present a risk.
- Use weather radio and alert for local weather watches and warnings.

Q-9 Mining

* Mining Categories

- There are three broad categories of mines in India.
  - Underground
  - Open cast
  - Surface
- While the first two categories represent accidents occurring inside the two types of mines, respectively, the third category represents mining-related accidents occurring above the surface in the vicinity of either type of mine.

* Important causes of Accidents in Mining

- Roof/side fall: accidental fall of or side at the time of excavation.
- Haulage: accident in the course of raising or lowering coal by tubs.
- Dumper: accidents in dumper-wheeled vehicles for carrying coal with tipper mechanism.
- Conveyer: accident in the user of belt or chain conveyers.
- Transport machinery: accidents associated with trucks and wagons.
- Explosive: accident in the course of using explosives.
- Electricity: accidents resulting out of the use of electricity.
- Dust/gas: accidents due to explosions of noxious gases, due to explosion of coal dust etc.
- Inundation: accidents due to sudden rushing in of water.
- Fall of object/person: accidents occurring due to sudden fall of an object or a person.

* Use of Technology to reduce accidents in Mining

- Design of system of support of roof in the development workings in underground mines by scientific support systems based on Rock Mass Rating (RMS) studies.
- Increased use of Roof Bolting / Roof Stitching methods of support using steel roof bolts / steel wire ropes.
- Introduction of modern drills like mechanized drilling machine to avoid exposure of support personnel to unsupported roof.

Q-10 Flood

- Floods are temporary inundation of large regions as a result of an increase in reservoirs or of rivers flooding their banks.
- A flood is an overflow of an expanse of water that submerges land.

* Natural Causes of Flood:

- Rapid excessive runoff from slopes to streams.
- Triggered by heavy rains.
- Promoted by steep slopes, low permeability soils, high orographic rainfall.
- Failure of dams and levees along rivers.
- Backing up of floos waters by sand barriers at river mouth.

* Human Causes of Flood:

- Undersized drainage betwork.
- Poor maintenance of drains and culverts.
- Indiscriminate garbage disposal in rivers.
- Poor or lack of maintenance of river channel.
- Low flow capacity of bridges.
- Inappropriate use of protective structure.
- Changes in land use leading to increased runoff.

*** Types of Flood:**

- Based on Duration
  - Slow-Onset floods
  - Rapid-Onset floods
  - Flash floods
- Based on Location
  - Coastal floods
  - Arroyos floods
  - River floods
  - Urban floods

*** Consequences and effects of Floods**

- Loss of human lives.
- Loss of property.
- Land degradation and erosion.
- The spawning grounds for fish and other wildlife habitants can become polluted or completely destroyed.

*** Precaution against Floods**

- First-aid kit should be kept ready.
- Radio, torch, candles and spare batteries should be kept at stock.
- Cattle should be removed to higher places.
- Medicines for snake bites and diarrhea should be kept at stock.
- Listen to radio or keep in touch with television for warning and advises.
- Adequate stock of fresh water, dry food articles, salt, sugar and kerosene should be kept handy.
- Only boiled water should be consumed and only freshly cook food should be eaten.

*** Flood Management**

- There are total three phase of flood management.

1) Floodplain management (before flood)
   - Preventive activities to reduce flood impacts.
   - Environmental management activities.
2) Flood response (during flood)
   - Operational activities.
   - Emergency management.
3) Flood recovery (after flood)
   - Assistance to get back to normal life.
   - Disaster relief.

Q-11 Cyclone

- Tropical storms, cyclones, hurricanes and typhoons, although named differently, describe the same disaster type.
- Essentially, these disaster types refer to a large scale closed circulation system in the atmosphere which combines low pressure and strong winds that rotate counter clockwise in the northern hemisphere and clockwise in the southern hemisphere.
- The system referred to as a “Cyclone” in the Indian Ocean and South Pacific, “Hurricane” in the Western Atlantic and Eastern Pacific and “Typhoon” in the Western Pacific.
- Hurricanes and Typhoons are the same storm types as “Tropical Cyclones”.
- Cyclones, hurricanes and typhoons can be predicted several days in advance.
- There are a number of structural characteristics common to all cyclones.
- A cyclone is a low pressure area. A cyclone’s center (eye), is the area of lowest atmospheric pressure in the region.

Q-12 Lightning

- Lightning is a bright flash of electricity produced by a thunderstorm.
- All thunderstorms produce lightning and are very dangerous.
- Causes of Lightning
- Lightning is an electric current. Within a thundercloud way up in the sky, many small frozen raindrops bump into each other as they move around in the air.
- All of those collisions create an electric charge.
- After a while, the whole cloud fills up with electrical charges.
- The positive charges from top of cloud and negative charges from bottom of cloud attract, that causes a positive charge to build up on the ground beneath the cloud.
- The grounds electrical charge concentrates around anything that sticks up.
- The charge coming up from these connects with a charge reaching down from clouds and zap lightning strikes is approximately 54000 °F. That is six times hotter than surface of the sun.

* Types of Lightning

1) Cloud-to-ground lightning
- It is most dangerous type of lightning.
- The cloud-to-ground lightning strike come from the negatively charged bottom of the cloud travelling to the positively charged ground below. It strikes on tall objects, like trees and buildings. It can cause fire and property damage.

2) Intracloud lightning
- It is the most common type of lightning. This occurs when there are both positive and negative charges within the same cloud. Usually, the process takes place within the cloud and looks like a bright flash of light which flickers.

3) Intercloud lightning
- It is less common. It occurs when there are positive and negative charges within different clouds and the strike travels in the air between them.

4) Forked lightning
- It appears as jagged lines of light. They can have several branches. It can be seen shooting from the clouds to the ground, from one cloud to another cloud, or from a cloud out into the air.

5) Sheet lightning
- It appears as flashes of light that seen to light up or illuminate entire clouds.
- Heat lightning
- It is a term used to describe lightning flashes that are too far away from you to hear the thunder.

Q-13 Avalanche

- The word avalanche is derived from the French word avalance meaning descent.
- An avalanche is a mass of snow, often mixed with ice and debris which travels down mountain sides, destroying all in its path.

* Causes of avalanche

- Overloading
- Temperature
- Slope angle
- Snow pack conditions
- Vibration

* Types of avalanche

@Powder Avalanche
- Often start from a single point and accumulates snow as it moves down the slope forming a snowball effect.
- This type is most common following heavy snowfall of one inch per hour or more.
- This type of avalanche can travel between 62 and 186 mph.

@Slab Avalanche

- Most common type of winter avalanche due to the build up fresh snow.
- A slab is a compact snow surface layer that can detach from a weaker snow layer underneath.
- The slab slips forward as a whole block or breaks into pieces.

@ Wet Avalanche

- Often occurs after a warm spell or during the spring snow.
- Snow becomes heavier as it begins to turn into water.
- Occurs frequently and are generally small and easier to predict than the other types.

Q-14 Drought

- According to meteorologists the rainfall deficiency during a long period over a large area is called a drought.

- Some times in Hindi language famine Akal and Anavrishty are also used for drought.

- Drought can also occur when ground water level is not within reach of agricultural communities.

*** Causes of Drought**

- Major cause of drought in India is scarcity of rain.

– But humans have interfered in the environment processes by their activities. People have filled up the natural resources like ponds and lakes. They have destroyed the vegetation cover.

- Vegetation cover impedes the flow of rainwater and force it to percolate in the ground.

- Humans have dug lakhs of tube wells and depleted the ground water reservoirs.

*** Effects of Consequences of Drought**

- Droughts cause scarcity of food and water.

- People die of hunger, malnutrition and epidemics.

- People are forced to migrate from their area of residence.

- Crops fail due to scarcity of water.

- Cattle fodder and water are not easily available.

- Farmers are deprived of their employment.

Q-15 Heat wave

- A heat wave is a period of abnormally high temperatures, more than the normal maximum temperature that occurs during the summer season in North-Western part of India.

- Heat waves typically occur between March and June, and in some rare cases even extend till July.

- The extreme temperatures and resultant atmospheric conditions adversely affect people living in these regions as they cause physiological stress, sometimes resulting in death.

*** Consequences and effects of Heat wave**

- Health Impacts of heat waves

- It involve dehydration, heat cramps, heat exhaustion and/or heat stroke.

– Heat cramps: Selling and Fainting generally accompanied by fever below 39°C.

- Heat exhaustion: Fatigue, weakness, headache, nausea, vomiting, muscle cramps etc.

- Heat stoke: Body temperature of 40°C or more along with coma.

Q-16 Forest Fire or Wild Fire

- Forest Fires are large fires which often start in wild land area.

*** Causes:**

- It is caused by Natural causes well as man made causes.

* **Natural causes**

- Many forest fires start from natural causes such as lightning which set trees on fire.
- However, rain extinguishes such fires without causing much damage.

* **Man made causes**

➢ Fire is caused when a source of fire like naked flame, cigarette or bidi, electric spark or any source of ignition comes into contact with inflammable material.

*** Classification of Forest Fires**

- Forest fires broadly be classified into three categories:

1) Natural or controlled forest fire.

2) Forest fires caused by heat generated in the litter and other biomes in summer through carelessness of people.

3) Forest fires purposely caused by local inhabitants.

*** Consequences and effects of Forest Fires**

- Loss of valuable timber resources.
- Degradation of catchment areas.
– Loss of biodiversity and extinction of plants and animals.
- Loss of wildlife habitat and depletion of wildlife.
- Loss of natural regeneration and reduction in forest cover.
- Global warming.
- Loss of carbon sink resource and increase in $\%CO_2$ in atmosphere.
- Ozon layer depletion.
- Soil erosion.

*** Precautions and measures of Forest Fires**

- Take fast initial attack measures.
- Vigorously follow up action.
- Introduce a forest fuel modification system at strategic points.
- Use firefighting resources wisely.
- Use forest protection systems

Q-17 Epidemics

- An epidemic is the rapid spread of infectious disease to a large number of people in a given population within a short period of time.

* **Causes:**

- Water supply
- Sanitation facilities
- Food
- Climate

*** The following factors favor occurrence of epidemics after disasters.**

- Temporary population settlements
- Pre-existent diseases in the population
- Ecological changes
- Resistance potential of the host
- Damage to public utility and interruption of public health services

*** Types of Epidemics**

* Point source

➢ An epidemic in which all cases are infected at the same time, usually from a single source or exposure.

* Continuous source

➢ An epidemic in which the causal agent (polluted drinking water, spoiled food) is infecting people who come into contact with it, over an extended period of time.

∗ Person-to-Person

- An epidemic in which the causal agent is transmitted from person to person, allowing the epidemic to propagate and spread as a function of Ro. E.g. is Swine flu.

*** Consequences and effects of Epidemics**

- It affect large number of individuals and can lead to complications including disabilities and death.
- On seeing the suffering and deaths especially within close relatives, psychological effects are also common during epidemics.
- Social and political disruption due to tension and law and order problems.
- Economic loss arising from lack of strength of cultivates.
- Scarcity of clean food and water leading to malnutrition.
- Worsening of poor sanitary conditions resulting in aggravation of epidemics.
- Pharmaceutical companies go for the research.

*** Prevention against Epidemics**

- Boil water before drinking.
- Keep food and water covered.
- Always wash hands thoroughly before taking any intakes.
- Drink fluid with sugar or glucose.
- Put alum or chlorine tablets in drinking water to disinfect.
- Eat something containing salts.
- Cook food thoroughly.

Q-18 Industrial Disaster

- It is the danger originating from technological or industrial accidents, dangerous procedures, infrastructure failures or certain human activities, which may cause the loss of life or injury, property damage, social and economic disruption or environmental degradation.
- Technological disasters are non-natural disastrous occurrence that include:

1) Accident release
   - Occurring during the production, transportation or handling of hazardous chemical substances.
2) Explosions
   - If the explosion is the cause of another disaster, the event will be classified as the resulting disaster.
3) Chemical explosion
   - Violent destruction caused by explosion of combustible material, nearly always of chemical origin.
4) Nuclear explosion/Radiation
   - Accidental release of radiation occurring in civil facilities, exceeding the internationally established safety levels.
5) Mine explosion
   - Accidents which occur when natural gas or coal dust reacts with the air.
6) Pollution
   - Degradation of one or more aspects in the environment by noxious industrial, chemical or biological wastes, form debris of man made products etc.
7) Acid rain
   - A washout of an excessive concentration of acidic compounds in the atmosphere.
8) Chemical pollution
   - A sudden pollution of water or air near industrial areas, leading to internal body disorders with permanent damage of the skin.
9) Atmospheric pollution
   - Contamination of the atmosphere by large quantities of gases, solids and radiation produced by the burning of natural and artificial fuels, chemicals and other industrial processes and nuclear explosions

Q-19 Disaster Risk Mapping

- Risk mapping is the process of establishing the spatial and temporal extent of risk (probability and consequences).
- The results of risk mapping are usually presented in the form of maps that show the magnitude and nature of the risk.
- Risk mapping requires combining maps of hazards, exposure, and vulnerability.
- The average frequency of occurrence and location of most extreme events can be determined with some degree of accuracy. While global maps of hazards, such as potential desertification, severe storms, and earthquake and volcanic activity, do exist, a more detailed approach is of more use to environmental health and disaster planners.

Q-20 IDNDR (International Decade for National Disaster Reduction)

- The United Nations General Assembly designated the 1990s as the International Decade for Natural Disaster Reduction (IDNDR).
- Its basic objective was to decrease the loss of life, property destruction and social and economic disruption caused by natural disasters, such as earthquakes, tsunamis, floods,landslides, volcanic eruption, droughts, and other disasters of natural origin.

* **What is needed for the success of IDNDR?**

- A large degree on the positive responses of outside agencies, engineers and scientists of all professions.
- Groups of people who are really devoted to take actions are needed.
- Team spirit among all nations, among engineers and scientists of different professions, and among all kinds of people working for disaster mitigation.

Technological Up gradation

- To share good practices and lessons learned to further disaster reduction within the context of attaining sustainable development, and to identify gaps and challenges.
- To increase awareness of the importance of disaster reduction policies.
- Public awareness and education, community participation.
- To increase the reliability and availability of appropriate disaster-related information to the public and disaster management agencies in all regions.

Q-21 Yokohama Strategy

* Strategy and plan of action – Introduction

- The World Conference on Natural Disaster Reduction, having met at Yokohama from 23 to 27 May 1994, recognizing the rapidly rising world-wide toll on human and economic losses due to natural disasters.

* Basis for Strategy

- Natural disasters continue to strike and increase in magnitude, complexity, frequency and economic impact.
- In this context the least developed countries, small island developing states and landlocked countries are the most vulnerable countries.
- In all countries the poor and socially disadvantaged groups suffer most from natural disasters and are least equipped to cope with them.

- Some patterns of consumption, production and development have the potential forincreasing the vulnerability to natural disasters.

* Strategy for the Year 2000 and beyond

- The world conference, based on adoption of the principles and the assessment of the progress accomplished during the first half of the Decade, has formulated a strategy for disaster reduction centered on the objective of saving human lives and protecting property.
- The strategy calls for an accelerated implementation of a Plan of Action to be developed from the following points:
- Development of a global culture of prevention as an essential component of an integrated approach to disaster reduction.

Q-22 Disaster Management in INDIA

- The Disaster, natural or man-made can destroy lives and properties on a very large scale, often pushing nations, in quest for progress, back by several decades.
- The Disaster, natural or man-made can destroy lives and properties on a very large scale, often pushing nations, in quest for progress, back by several decades.

*** Paradigm Shift in Disaster Management**

- The Disaster, natural or man-made can destroy lives and properties on a very large scale, often pushing nations, in quest for progress, back by several decades.
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*** Objectives**

- Mitigation or reduction of risk of any disaster or its severity or consequences.
- Capacity building including research and knowledge management.
- Preparedness to deal with any disaster.
- Prompt response to any threatening disaster situation or disaster.
- Assessing the severity or magnitude of effects of any disaster.
- Evacuation, rescue and relief. Rehabilitation and reconstruction.

Q-23 National Disaster Management Authority (NDMA)

- The NDMA, as the apex body for disaster management, is headed by the Prime Minister and has the responsibility for laying down policies, plans and guidelines for DM.
- The guidelines will assist the Central Ministries, Departments and States to formulate their respective DM plans.
- It will approve the National Disaster Management and DM plans of the Central Ministries/Departments.
- Central ministries/departments and State Governments will extend necessary cooperation and assistance to NDMA for carrying out its mandate.
- The general superintendence, direction and control of National Disaster Response Force (NDRF) are vested in and will be exercised by the NDMA.
- The National Institute of Disaster Management (NIDM) works within the framework of brad policies and guidelines laid down by NDMA.

Q-24 National Executive Committee (NEC)

- The NEC, comprises the Union Home Secretary as the Chairperson, and the Secretaries to the GOI in the Ministries/Departments of Agriculture, Atomic Energy, Defense, Drinking Water supply, Environment and Forests, Finance, Health, Power, Rural Development, Science and Technology, Space, Telecommunications, Urban Development, Water Resources and the Chief of the Integrated Defense Staff as members.
- NDMA will be special invitees to the meetings of the NEC.
- The NEC is the executive committee of the NDMA, and is mandated to assist the NDMA in the discharge of its functions and also ensure compliances of the directions issued by the Central Government.
- NEC is to coordinate the response in the event of any threatening disaster situation or disaster.
- NEC will prepare the National Plan for Disaster Management based on the National Policy on Disaster Management.

Q-25. National Disaster Organizations

-National Institute of Disaster Management (NIDM)

- The NIDM, in partnership with other research institutions has capacity development as one of its major responsibilities, along with training, research, documentation and development of a national level information base.
- It will network with other knowledge-based institutions, and function within the broad policies and guidelines laid down by the NDMA.
- It will organize training of trainers, DM officials and other stakeholders.
- The NIDM will strive to emerge as a 'Centre of Excellence' in the field of Disaster Management.

Q-26 National Disaster Response Force (NDRF)

- For the purpose of specialized response to a threatening disaster situation or disasters/emergencies both natural and man-made such as those of Chemical, Biological,Radiological and Nuclear origin, the Act has mandated the constitution of a National Disaster Response Force (NDRF).
- The general superintendence, direction and control of this force shall be vested in and exercised by the NDMA and the command and supervision of the Force shall vest in an officer to be appointed by the Central Government as the Director General of Civil Defense and National Disaster Response Force.
- Presently, the NDRF comprises eight battalions.

Q-27 Role of Information Technology

- For correct decision making at any stage of natural disasters, from prediction to reconstruction, a considerable amount of data and information is necessary.
- Experience has proved that information technology facilitates the receiving, classifying, analyzing, and dissemination of information for appropriate decision making.
- The main data and information critical for an efficient and robust disaster management system are those made available from :-
- Observatory stations
- Satellites observed
- Centre-to-centre
- Classified experiences
- Research results

- Training contents
- Reports and
- news

Q-28 Role of Remote Sensing and GIS

-Remote sensing is an investigative technique that uses a recording instrument or device to measure or acquire information on a distant object or phenomenon with which it is not in physical or intimate contact.

-The technique is used for accumulating vital information on the environment.

- It comprises Aerial Remote Sensing, which is the process of recording information such as photographs and images from sensors on aircrafts and satellite remote sensing.

- Potential applications of remote sensing in disaster management include the following

- To map the variations in terrain properties, such as vegetation, water, and geology, both in space and time.
- Helping to locate the area of a natural disaster and monitor its growing proportions while the forces of disaster are in full swing.
- Monitoring the disaster event which provides, in turn, a quantitative base for relief operations. Such assessment can be used to map the new scenario and update the database used for the reconstruction of the crisis area.

*** Advantages of Remote Sensing**

- Saves time
- Users of the technology do not have to be in direct contact with danger zones.
- Shows image of very large areas of land or space.
- Detect features at wavelengths not visible to the human eye.
- Data can be regularly and routinely acquired and archived.
- The most cost-effective dataset for monitoring change over large areas.
- Can assist with damage assessment monitoring.

*** Limitation of Remote Sensing**

- It can be costly to build and operate a remote sensing system.
- Data can be difficult to interpret and may require expert skills.
- Resolution is often coarse.
- Small size activities cannot be delineated on remote sensing imagery or through aerial photography.

Q-29 Geographical Information System (GIS)

- The GIS can be loosely defined as a system of hardware and software used for measuring, storing, retrieving, mapping, monitoring, modeling, and analyzing a variety of data types related to geographic and natural phenomena.
- In other words, GIS is a computer-based system capable of integrating, storing, editing, analyzing, sharing, and displaying.
- The **most common applications of GIS** in disaster management are the following:

- GIS provides a versatile platform for decision support.
- Hazard mapping for risk assessment.
- GIS is used as a tool for the planning of evacuation routes.
- GIS is used to organize the damage information.
- GIS facilitates the calculation of emergency response times for emergency planners in the event of a natural disaster.
- GIS based database will ensure the mobilization of the necessary resources.

*** Advantages of GIS**

- GIS has the ability to represent spatial information over a wide geographic area.
- GIS effectively analyzes, collects, manages and distributes up-to-date information.
- GIS is versatile and easy to use.
- Information from GIS can be easily tabulated.

*** Limitation of GIS**

- Major impacts on life of people, economy and environment.
- GIS being a technological tool can be complex and a bit difficult to grasp initially.
- Large amounts of information is usually required to get useful output from the system.
- The decision making process may be stalled during an emergency.

*** INTEGRATION OF REMOTE SENSING AND GIS**

- The integration of satellite data into Geographic Information System (GIS) is one of the great idea that focus on the rapid acceptance of GIS technology in to the geo information oriented applications in operational environments.
- In remote sensing the data gathered is very large. So, comprehensive digital processing system is used to manage, manipulate and analyze these data.
- GIS is used for spatial data analysis linked with attribute data. GIS is widely used for handling geographic data and performing analysis on those data for a number of earth related disciplines.

-The process of integrating remote sensing data into a GIS includes following analytical procedures:

- Data acquisition
- Data processing
- Data analysis
- Data conversion
- Error assessment

Q-30 Global Positioning System (GPS)

- The Global Positioning System (GPS) serves as a facilitating technology in addressing these needs by helping the users, at any point on or near the earth's surface, to obtain instantaneous three-dimensional coordinates of their location.

• Major applications of GPS include:

- Pinpointing the location of damage sites and floodplains.
- Playing an increasingly prominent role in helping scientists to anticipate earthquakes in earthquake prone areas.
- Meteorologists responsible for storm tracking and flood prediction also rely on GPS.
- GPS has become an integral part of modern emergency response systems.
- GPS has given managers a quantum leap forward in the efficient operation of their emergency response teams.
- Incorporation of GPS in mobile phones places an emergency location capability in the hands of everyday users.